

Experimental study of monazite solubility in haplogranitic melts: a new model for peraluminous and peralkaline melts

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Abstract: Monazite is a common accessory mineral in peraluminous, metaluminous and peralkaline granitic/rhyolitic rocks. Considering the importance of monazite in geochemical and geochronological studies, a monazite solubility model that can be applied to a wide compositional range of magmas is desirable. To accomplish this, monazite solubility experiments were performed at atmospheric (1400 °C) and crustal pressures (1–3 kbar, 720–850 °C, H₂O-saturated), using haplogranitic compositions ranging from peraluminous to peralkaline, doped with synthetic pure LaPO₄. The concentrations of La in the melts increase sharply with increasing temperature and peralkalinity of the melt. We combined our new data with those of previous studies to describe the solubility of monazite in peralkaline to peraluminous melts. Our new monazite saturation model, which incorporates temperature, pressure, water content, melt and monazite composition is given by: $\ln \Sigma \text{LREE} = 12.77(\pm 0.49) + 1.52(\pm 0.15)M + 0.44(\pm 0.10)(\text{H}_2\text{O})^{0.5} - 9934(\pm 632)/T - 36.79(\pm 6.15)P/T + \ln X_{\text{mnz}}^{\text{LREE}}$ where ΣLREE is the sum of the concentrations of La to Sm in monazite-saturated melt, in ppm, M is a dimensionless compositional parameter $(\text{Na} + \text{K} + 2\text{Ca}) \times \text{Al}^{-1} \times (\text{Al} + \text{Si})^{-1}$, similar to the compositional parameter used in an earlier model by Montel; Na, K, Ca, Al, Si are in moles, H₂O is water content in weight percent, T is the temperature in K, P is the pressure in kbar, and $X_{\text{mnz}}^{\text{LREE}}$ is the mole fraction of LREE in monazite $\text{LREE}_{\text{mnz}}/(\text{LREE}_{\text{mnz}} + \text{Y}_{\text{mnz}} + \text{Th}_{\text{mnz}} + \text{U}_{\text{mnz}})$. This model reproduces 76% and > 95% of the data to within uncertainties of $\pm 10\%$ and $\pm 20\%$, respectively. It may be applied to felsic melts poor in CaO + FeO + MgO (< 3 wt%) from peraluminous to peralkaline compositions.

Key-words: monazite; solubility; melt composition; agpaitic index; peralkaline; peraluminous; haplogranite; model; experimental petrology.

1. Introduction

Monazite is a light rare earth element-bearing (LREE) phosphate occurring as an accessory mineral in a wide variety of rock types ranging from peraluminous to peralkaline as well as in carbonatites and kimberlites, and some metamorphic rocks (e.g. Lyakhovich & Barinskii, 1961; Overstreet, 1967; Bea, 1996; Jones *et al.*, 1996; Spear & Pyle, 2010).

Other than providing a major repository for LREE, monazite can also host significant amounts of Y, Th, and U and thus is often employed in Th-U-Pb-dating of crustal rocks. It is also an important sink for LREE, U, Th in geochemical modelling of trace elements during the crystallization of magma, or during crustal melting (Miller & Mittlefehldt, 1982; Rapp *et al.*, 1987; Seydoux *et al.*, 2002; Hermann & Rubatto, 2009; Skora & Blundy, 2010; Poletti *et al.*, 2017) and is an economic source of

strategic elements (Chakhmouradian & Zaitsev, 2012; Chakhmouradian & Wall, 2012).

For all of the above reasons, knowledge of the solubility behaviour of monazite in magmatic processes plays a central role in petrogenetic models and thus it has been studied since the late 1980s. Experimental studies investigated monazite saturation in both model and natural systems in a wide range of P/T conditions (1–50 kbar/750–1400 °C), H₂O contents (1–29 wt%), and experiments were performed using melts of different compositions which can be expressed by an agpaitic index value ($\text{AI} = (\text{Na} + \text{K})/\text{Al}$ in molar units) from 0.45 to 2.3. Temperature and melt peralkalinity ($\text{AI} > 1$) in granitic systems seem to have a strong positive effect on monazite solubility, whereas in peraluminous systems ($\text{AI} < 1$) the effect of melt composition seem to be weaker. Pressure and water content in the melt seem to show, respectively, little effect (Montel, 1986, 1993; Rapp & Watson, 1986; Rapp *et al.*, 1987;

Hermann & Rubatto, 2009; Xing *et al.*, 2013; Skora & Blundy, 2010, 2012) or a negative effect on monazite solubility (Stepanov *et al.*, 2012), whereas fluorine has no documented effect (Keppler, 1993; Duc-Tin & Keppler, 2015).

Several empirical equations linking monazite saturation to the physical and chemical parameters influencing its solubility have been proposed under the general form:

$$\ln \text{LREE}_{\text{melt}} = a + bT^{-1} + cPT^{-1} + dM + e(\text{H}_2\text{O})^{0.5} + \ln X_{\text{mnz}}^{\text{LREE}} \quad (1)$$

where $\text{LREE}_{\text{melt}}$ is the sum of La–Sm in ppm; a , b , c , and d are constants derived from the experimental data by linear multicomponent regression; T is temperature in K, P is pressure in kbar; M is a compositional parameter $[(\text{Na} + \text{K} + \text{Li} + 2\text{Ca}) \times \text{Al}^{-1} \times (\text{Al} + \text{Si})^{-1}]$ expressing the effect of melt composition in atomic fraction; H_2O is in wt%; $X_{\text{mnz}}^{\text{LREE}}$ is the mole fraction of LREE in monazite, assumed to be proportional to the activity of LREE in monazite (Stepanov *et al.*, 2012). Rapp *et al.* (1987) proposed a simpler empirical equation which links monazite solubility to temperature from experiments by Rapp & Watson (1986):

$$\ln \Sigma \text{LREE} = 18.5 - 14160T^{-1} \quad (2)$$

where ΣLREE is the sum of LREE (La–Sm) dissolved in the melt (in ppm). However, Eq. (2) does not take into account the effect of melt and monazite composition. Subsequently, Montel (1993) proposed an empirical equation for monazite solubility in peraluminous to peralkaline silicic compositions using data from three experimental studies (Montel, 1986, 1993; Rapp & Watson, 1986) and taking into account the effects of water content, melt and monazite composition:

$$\ln(\text{REE}_t/X_{\text{REEPO}_4}) = 9.50 + 2.34M + 0.3879(\text{H}_2\text{O})^{0.5} - 13318T^{-1} \quad (3)$$

where $\text{REE}_t = \Sigma \text{REE}_i$ (ppm)/atomic weight (g mol^{-1}) and X_{REEPO_4} is the sum of the mole fractions of REE-phosphates in monazite. More recently, Stepanov *et al.* (2012) combined 57 experimental runs with peraluminous to metaluminous compositions to calibrate a new equation that can predict monazite solubility as a function of pressure, temperature, melt water content and monazite composition:

$$\ln \Sigma \text{LREE} = 16.16(\pm 0.3) + 0.23(\pm 0.07)(\text{H}_2\text{O})^{0.5} - 11494(\pm 410)T^{-1} - 19.4(\pm 4)PT^{-1} + \ln X_{\text{mnz}}^{\text{LREE}} \quad (4)$$

Given the limited compositional range $[0.85 < \text{Al}/(2\text{Ca} + \text{Na} + \text{K}) < 1.2]$ on which Eq. (4) was calibrated, it should be used only for peraluminous to metaluminous compositions.

Considering these previous studies, we set out to extend the experimental database for monazite solubility using a model system $\text{SiO}_2\text{--Al}_2\text{O}_3\text{--Na}_2\text{O--K}_2\text{O}$ both for peraluminous and peralkaline compositions. In this article, we report new experimental results on La-monazite saturation at high temperatures and atmospheric (anhydrous melts) and crustal pressures (hydrous melts) in peraluminous and peralkaline melts, with the goal of obtaining a refined model for monazite solubility which may be used for peraluminous to peralkaline magma compositions.

2. Experimental and analytical techniques

2.1. Starting materials

Synthetic pure La-monazite and a series of haplogranitic glasses (HPG8), with Al ranging from 0.84 to 1.93, used as starting compositions (Table 1) for the solubility experiments were prepared at the Ludwig Maximilians Universität in Munich (Germany). The pure monazite was prepared by mixing La_2O_3 and $\text{NH}_4\text{H}_2\text{PO}_4$ in stoichiometric proportions to yield LaPO_4 and the mixture was heated in a Pt crucible, in air, for 15 h at 1015 °C. X-ray diffraction analyses confirmed the purity of La-monazite. The haplogranitic starting glasses were prepared by mixing desired proportions of oxides (Si, Al) and carbonates (K, Na) in an agate mortar under ethanol and then fusing the mixture at 1600 °C for several hours. The Pt crucible was removed and air-cooled to room temperature. The resulting haplogranitic glasses were bubble-rich and inhomogeneous. To remove the bubbles and promote homogenization, the fused batches were loaded into a viscometer furnace, reheated at 1650 °C while a stirring spindle ($\text{Pt}_{80}\text{Rh}_{20}$) was introduced into the silicate liquids. The silicate liquids were stirred for up to two weeks, until periodical inspection of the spindle indicated that the melts were homogeneous and bubble-free. Bubble-free melts were cooled slowly to <800 °C and then air-cooled to room temperature. All products were ground again in the agate mortar under ethanol until fine grain powders (~10–20 μm) were obtained.

In the HPG8-based peraluminous and peralkaline compositions we keep the same molar Na/K ratio as in pure HPG8 composition, yet with different $(\text{Na} + \text{K})/\text{Al}$, in order to investigate the role of peralkalinity without changing the melt Na/K ratio.

2.2. Hydrothermal experiments and high- T experiments

Gold capsules (1.8 cm in length, inner diameter 2.5 mm, outer diameter 3 mm) were used as sample containers. First, a few microliters of distilled and deionized water were inserted with a microsyringe, then ~40 mg of the haplogranitic starting material, doped with ~5 wt% of synthetic La-monazite, were added. Liquid water was added in the amount required to ensure water-saturated conditions for all experiments, according to the solubility data of Holtz *et al.* (1992). Capsules were weighed after each addition of material and then were crimped and sealed by arc-welding.

Table 1. Average electron microprobe analyses of glasses used as starting materials (wt%).

Glass	Na ₂ O	K ₂ O	Al ₂ O ₃	SiO ₂	Total	<i>n</i>	(Na + K)/Al
HPG8-0.8	3.87 (31)	3.80 (19)	12.54 (38)	78.91 (31)	99.12	10	0.84
HPG8-0.9	4.98 (52)	4.12 (15)	13.58 (9)	76.66 (54)	99.34	8	0.93
HPG8-1.5	6.85 (41)	5.39 (35)	11.80 (62)	75.40 (99)	99.44	8	1.45
HPG8-2	8.37 (70)	7.29 (39)	11.21 (17)	72.63 (20)	99.50	8	1.93

Note: Values in parentheses are one standard deviation of *n* replicate analyses.

The welded capsules were heated overnight at 110 °C and weighed again to check for leaks.

Water-saturated experiments were performed in water-pressurized vertical cold seal pressure vessels (CSPV, Nimonic 105) at the Geology Division of Camerino University (Italy). The intrinsic redox conditions of the apparatus are close to NNO + 0.8 (Di Matteo *et al.*, 2004; Fabbrizio *et al.*, 2006; Fabbrizio & Carroll, 2008). A total of 43 experiments were carried out (Table 2). Experimental temperatures varied from 720 to 850 °C with pressure from 1 to 3 kbar; run durations were between 5 and 22 days depending on temperature, with longer durations at lower *T* and *P* conditions. The runs were terminated by switching off the power supply and the samples were quenched isobarically in a stream of compressed air, cooling the sample to near room temperature in ~10 min. In order to avoid exsolution of water during the quenching process, the pressure was kept constant using a hand-operated pressure intensifier. Temperature was measured near the sample position with a sheathed K-type thermocouple placed in a borehole at the base of the autoclave (accuracy is estimated as ±5 °C). Pressure was monitored by a high-pressure transducer or Bourdon-tube pressure gauges, considered accurate to ±0.025 kbar.

After each experiment, the capsules were weighed to check for leaks and opened; only samples showing no weight changes were accepted as successful. For each charge one half of the run product was mounted in epoxy and polished, some as normal thin sections, and others as epoxy-embedded thick sections. Monazite can easily be identified optically in dissolution experiments since it is a highly birefringent crystal (thin sections) and it has strong reflectance in contrast with silicate glass in reflected light (thick sections). Run products contained only glass, bubbles and 20–50 µm sized monazite crystals.

Seven high-temperature experiments (Table 2) were carried out at the Department of Earth and Environmental Sciences of the Ludwig Maximilian University of Munich (LMU, Germany) at atmospheric pressure and 1400 °C with duration of 12–36 h using Pt capsules as sample containers. The samples were quenched by removing them from the furnace and immersion in water. The purpose of the experiments at 1400 °C was to examine the effect of temperature on monazite solubility. All charges were mounted in epoxy and polished before carbon coating for electron-microprobe analysis.

2.3. Electron-microprobe analysis

Major and minor elements were analyzed using a Cameca SX100 electron microprobe at the CNR Institute for Geosciences and Earth Resources, Padova. Crystals were

analyzed with a 1 µm spot while a 10 µm beam size, to minimize alkali loss, was used for analyses of the quenched glasses. Analytical conditions were 20 kV accelerating potential, 10 nA beam current and counting times of 10 s on peak (half on background) for Si, Al, and alkalis (always analyzed first), and 30 nA and 20 s for La and P. Standard materials used for calibration were quartz for Si, corundum for Al, aegirine for Na, sanidine for K, apatite for P, and end-member synthetic phosphate glass (LaP₅O₁₄) for La. A PRZ correction procedure (Reed, 2005) was used to convert specimen/standard intensity ratios into concentrations. Some additional analyses of glass compositions (several anhydrous, 1 atm experiments) were also obtained using the electron microprobe at the University of Milan, using analysis conditions similar to those used for the Padova electron microprobe. Analyses of the same samples with the two microprobes gave results identical within analytical uncertainties.

3. Results

3.1. Dissolution mechanisms of La-monazite

Four series of experiments were carried out using starting materials with AI of 0.84, 0.93, 1.45 and 1.93 to study the effect of varying melt compositions on the solubility of pure La-monazite. Run products were hydrous glasses containing bubbles and monazite crystals. Small bubbles (>~5 µm) result from exsolution during quenching (rare) and larger bubbles are from fluid equilibrated with melt. Monazite crystal sizes are predominantly 20–50 µm diameter with near equant shapes. Table 3 compiles average microprobe data of the run product glasses and solubility data. The chemical homogeneity of the glasses is indicated by the relatively small standard deviations of the multiple analysis points (Table 1). The H₂O contents of glasses were estimated using the model of Papale *et al.* (2006).

The high-pressure data in Table 3 show that La concentration in peralkaline melts increases with increasing phosphorous content in the melt (Fig. 1). The positive relationship between La and P concentrations is likely a reflection of approximately stoichiometric monazite dissolution related to the presence of molecular species in the silicate melt.

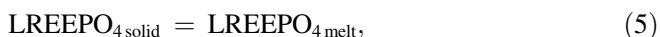
Indeed, it has been proposed that dissolution of monazite in silicate melts can occur as dissolution of molecular species which upon dissolution may dissociate into ionic species (Skora & Blundy, 2012; Duc-Tin & Keppler, 2015). In this last case, the REE and P concentrations in the melt

Table 2. Experimental run conditions.

Run	Starting glass	LaPO ₄ (wt%)	H ₂ O (wt%)	T (°C)	P (kbar)	Time (d)
CHG-22	HPG8-0.8	5	7	800	3	7
CHG-25	HPG8-0.8	5	7	850	2	7
CHG-1	HPG8-0.8	5	7	800	2	10
CHG-16	HPG8-0.8	5	7	850	1.5	7
CHG-13	HPG8-0.8	5	7	800	1.5	7
CHG-10	HPG8-0.8	5	7	780	1.5	5
CHG-19	HPG8-0.8	5	7	750	1.5	12
CHG-28	HPG8-0.8	5	7	850	1	7
CHG-4	HPG8-0.8	5	5	800	1	17
CHG-9	HPG8-0.8	5	5	800	1	7
CHT-1	HPG8-0.8	2.5	—	1400	0.001	0.5
CHT-4	HPG8-0.8	2.5	—	1400	0.001	1
CHT-7	HPG8-0.8	2.5	—	1400	0.001	1.5
CMH-1	HPG8-0.9	5	7	850	2	12
CMH-4	HPG8-0.9	8	7	800	2	22
CMH-6	HPG8-0.9	8	7	780	2	8
CMH-5	HPG8-0.9	8	7	760	2	8
CMH-9	HPG8-0.9	9	7	740	2	22
CMH-8	HPG8-0.9	9	7	720	2	22
CMH-3	HPG8-0.9	5	5	850	1.5	12
CMH-2	HPG8-0.9	5	5	850	1	12
CHG-23	HPG8-1.5	5	7	800	3	7
CHG-26	HPG8-1.5	5	7	850	2	7
CHG-2	HPG8-1.5	5	7	800	2	10
CHG-17	HPG8-1.5	5	7	850	1.5	7
CHG-14	HPG8-1.5	5	7	800	1.5	7
CHG-11	HPG8-1.5	5	7	780	1.5	5
CHG-20	HPG8-1.5	5	7	750	1.5	12
CHG-29	HPG8-1.5	5	7	850	1	7
CHG-7	HPG8-1.5	5	5	800	1	17
CHT-2	HPG8-1.5	2.5	—	1400	0.001	0.5
CHT-5	HPG8-1.5	2.5	—	1400	0.001	1
CHT-8	HPG8-1.5	2.5	—	1400	0.001	1.5
CHG-24	HPG8-2	5	7	800	3	7
CHG-27	HPG8-2	5	7	850	2	7
CHG-3	HPG8-2	5	7	800	2	10
CHG-18	HPG8-2	5	7	850	1.5	7
CHG-15	HPG8-2	5	7	800	1.5	7
CHG-12	HPG8-2	5	7	780	1.5	5
CHG-21	HPG8-2	5	7	750	1.5	12
CHG-30	HPG8-2	5	7	850	1	7
CHG-8	HPG8-2	5	5	800	1	17
CHT-3	HPG8-2	2.5	—	1400	0.001	0.5

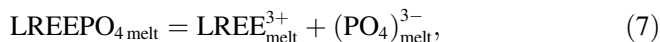
Note: The suffix of HPG8 indicates the (Na + K)/Al molar ratio of the starting glass.

must vary inversely (Skora & Blundy, 2012; Duc-Tin & Keppler, 2015), but it appears that at low pressures, for starting materials with approximately stoichiometric La/P abundances (as in our experiments), monazite dissolution is mainly as an associated LREEPO₄ complex (Montel, 1993; Duc-Tin, 2007; Skora & Blundy, 2012). The equilibrium between solid monazite and its molecule species in the melt is controlled by the reaction:



$$K_{(1)} = a(\text{LREEPO}_{4\text{melt}})/a(\text{LREEPO}_{4\text{solid}}), \quad (6)$$

where K is the equilibrium constant and a is activity, and upon dissolution it may dissociate into their ionic species:



$$K_{(2)} = (a(\text{LREE}_{\text{melt}}^{3+}) a((\text{PO}_4)_{\text{melt}}^{3-})) / a(\text{LREEPO}_{4\text{melt}}). \quad (8)$$

Substituting Eq. (6) for pure phosphates of one single rare earth, *i.e.* $a(\text{LREEPO}_{4\text{solid}}) = 1$, in Eq. (8) yields

$$K_{(2)} = 1/K_{(1)} a(\text{LREE}_{\text{melt}}^{3+}) a((\text{PO}_4)_{\text{melt}}^{3-}). \quad (9)$$

Then, approximating activities by concentrations (denoted by components in square brackets) gives

$$\begin{aligned} [\text{LREE}_{\text{melt, total}}] &= [\text{LREEPO}_{4\text{melt}}] + [\text{LREE}_{\text{melt}}^{3+}] \\ &= K_{(1)} + K_{(1)} K_{(2)} [(\text{PO}_4)_{\text{melt}}^{3-}]^{-1}. \end{aligned} \quad (10)$$

Therefore, if the dissociation reaction (7) is involved in driving the La-monazite solubility, the LREE concentration in the melt plotted against the inverse of the P concentration should describe a linear trend. In this study the melt initially was nominally P- and La-free, but exhibit a concomitant increase of both elements in the run products (Fig. 1), with La/P slope close to the ideal stoichiometric value (4.48). This observation is in good agreement with the idea that the dissolution of molecular species is the dominant process in controlling the La-monazite solubility at crustal pressures, in line with the observations of Montel (1993) and Duc-Tin (2007), and thus the concentration of dissolved La should suffice to characterize monazite solubility (*cf.* Table 3).

3.2. The effect of the agpaitic index on monazite solubility

A comparison of the measured solubility for pure La-monazite coexisting with the haplogranitic melts at the different run temperatures is shown in Fig. 2 as a function of the agpaitic index (AI = (Na + K)/Al in molar units). The general trend is that peraluminous melts show solubility that are systematically lower than those of peralkaline melts at comparable temperatures; *e.g.* at 1400 °C La concentrations decrease from 1.35 to 1.5 wt% in peralkaline melts with AI in the range 1.22–1.69 to 0.55–0.7 wt% in peraluminous melts with AI in the range 0.86–0.89. In detail, for the peralkaline melts (AI > 1) the solubility of monazite increases with increasing agpaitic index whereas, for the peraluminous melts (AI < 1), the solubility of monazite seems to increase slightly with decreasing AI. The solubility increase for peralkaline compositions and its small dependence on the agpaitic index for peraluminous compositions has been observed also by Duc-Tin & Keppler (2015).

As noted above, our data suggest that La-monazite dissolves as molecular species and the solubility increase in

Table 3. Average electron microprobe analyses of run products (wt%).

Run	Na ₂ O	K ₂ O	Al ₂ O ₃	SiO ₂	P ₂ O ₅	La ₂ O ₃	Total	H ₂ O	n	Al	P (ppm)	La (ppm)
CHG-22	3.32 (19)	3.48 (12)	12.07 (16)	74.33 (41)	0.12 (10)	0.17 (24)	93.50	6.50	17	0.76	565	1567
CHG-25	3.41 (34)	3.61 (13)	12.26 (21)	74.82 (63)	0.13 (13)	0.15 (26)	94.39	5.61	15	0.78	595	1360
CHG-1	3.09 (69)	3.54 (10)	11.99 (24)	75.62 (72)	0.11 (11)	0.15 (31)	94.53	5.47	7	0.74	524	1373
CHG-16	3.37 (42)	3.63 (11)	12.23 (17)	75.47 (58)	0.16 (9)	0.23 (25)	95.11	4.89	13	0.77	724	2042
CHG-13	3.19 (57)	3.67 (16)	12.21(20)	75.66 (66)	0.18 (7)	0.27 (17)	95.20	4.80	18	0.75	821	2428
CHG-10	3.01 (43)	3.58 (15)	12.25 (14)	76.03 (78)	0.15 (18)	0.20 (35)	95.25	4.75	7	0.72	699	1826
CHG-19	3.34 (19)	3.60 (14)	12.17 (30)	75.83 (44)	0.10 (7)	0.13 (17)	95.28	4.72	17	0.78	470	1121
CHG-28	3.31 (38)	3.66 (14)	12.26 (16)	76.44 (43)	0.14 (11)	0.20 (24)	96.02	3.98	14	0.77	628	1749
CHG-4	3.12 (51)	3.21 (22)	16.82 (15)	72.27 (21)	0.22 (10)	0.34 (25)	96.02	3.98	7	0.51	997	3049
CHG-9	3.78 (18)	3.78 (12)	12.18 (11)	75.92 (37)	0.13 (9)	0.16 (17)	95.98	4.02	6	0.85	594	1419
CHT-1	4.18 (19)	3.96 (10)	12.85 (15)	77.96 (37)	0.37 (7)	0.65 (3)	97.79	–	18	0.87	1627	5518
CHT-4	4.06 (27)	3.84 (12)	12.57 (12)	78.22 (41)	0.47 (9)	0.81 (3)	98.42	–	16	0.86	2063	6875
CHT-7	4.47 (31)	3.54 (14)	12.54 (14)	78.23 (36)	0.38 (10)	0.82 (5)	98.90	–	20	0.89	1680	7030
CMH-1	4.06 (50)	4.09 (36)	12.00 (61)	74.02 (99)	0.11 (3)	0.02 (5)	94.31	5.69	18	0.93	213	491
CMH-4	4.19 (19)	4.05 (17)	12.36 (32)	73.67 (40)	0.16 (3)	0.06 (6)	94.50	5.50	9	0.91	719	551
CMH-6	3.22 (50)	4.16 (17)	12.42 (46)	74.69 (81)	0.12 (3)	0.06 (10)	94.68	5.32	11	0.79	554	546
CMH-5	3.04 (26)	4.25 (10)	12.95 (21)	74.44 (44)	0.13 (5)	0.03 (2)	94.85	5.15	5	0.74	615	247
CMH-9	4.02 (73)	3.87 (18)	12.51 (42)	74.05 (83)	0.10 (5)	0.02 (8)	94.76	5.24	9	0.89	452	208
CMH-8	4.45 (27)	3.91 (19)	12.32 (80)	73.97 (107)	0.11 (3)	0.02 (5)	94.79	5.21	10	0.94	486	179
CMH-3	3.73 (33)	4.22 (14)	12.28 (30)	74.56 (38)	0.22 (13)	0.09 (6)	95.10	4.90	8	0.87	987	764
CMH-2	4.07 (73)	4.22 (18)	12.56 (42)	74.90 (83)	0.17 (5)	0.04 (8)	95.99	4.01	9	0.89	382	789
CHG-23	5.25 (19)	5.40 (16)	11.36 (26)	70.72 (34)	0.14 (8)	0.26 (20)	93.16	6.84	8	1.27	672	2397
CHG-26	5.30 (35)	5.51 (20)	11.48 (24)	71.23 (47)	0.18 (6)	0.37 (15)	94.07	5.93	10	1.28	814	3321
CHG-2	5.15 (42)	5.79 (21)	11.37 (29)	71.27 (75)	0.24 (21)	0.37 (17)	94.21	5.79	8	1.30	1127	3310
CHG-17	5.55 (19)	5.64 (15)	11.65 (25)	71.53 (38)	0.15 (6)	0.24 (14)	94.77	5.23	9	1.31	670	2189
CHG-14	5.38 (36)	5.76 (20)	11.44 (19)	71.76 (47)	0.16 (8)	0.31 (19)	94.83	5.17	8	1.32	741	2752
CHG-11	5.23 (19)	5.60 (5)	11.42 (42)	72.21 (49)	0.16 (1)	0.27 (17)	94.90	5.10	5	1.28	717	2444
CHG-20	5.06 (68)	5.60 (14)	11.42 (23)	72.44 (63)	0.17 (8)	0.27 (16)	94.99	5.01	10	1.26	792	2379
CHG-29	5.77 (24)	5.60 (15)	11.71 (23)	72.09 (21)	0.18 (4)	0.32 (13)	95.68	4.32	6	1.33	804	2868
CHG-7	5.60 (19)	5.64 (12)	11.64 (22)	72.32 (38)	0.18 (3)	0.28 (13)	95.68	4.32	5	1.32	822	2478
CHT-2	6.55 (22)	5.97 (10)	12.25 (20)	72.85 (19)	0.78 (3)	1.58 (9)	97.84	–	17	1.41	3388	13 446
CHT-5	5.71 (15)	4.81 (15)	11.99 (17)	75.18 (21)	0.69 (4)	1.61 (7)	98.46	–	10	1.22	2998	13 723
CHT-8	6.25 (17)	5.25 (22)	11.84 (23)	74.34 (15)	0.63 (4)	1.67 (6)	99.33	–	27	1.35	2759	14 246
CHG-24	5.55 (22)	7.03 (23)	11.43 (21)	68.55 (27)	0.16 (5)	0.31 (11)	93.04	6.96	13	1.46	748	2812
CHG-27	6.79 (19)	7.06 (16)	11.37 (21)	67.65 (36)	0.27 (2)	0.51 (6)	93.67	6.33	14	1.65	1281	4658
CHG-3	5.11 (36)	7.43 (27)	11.59 (27)	69.27 (14)	0.27 (6)	0.50 (16)	94.18	5.82	5	1.42	1230	4542
CHG-18	6.94 (4)	7.08 (10)	11.54 (25)	68.02 (36)	0.28 (3)	0.51 (5)	94.40	5.60	9	1.64	1317	4605
CHG-15	5.76 (31)	7.19 (15)	11.70 (16)	69.41 (25)	0.24 (4)	0.41 (10)	94.73	5.27	7	1.47	1127	3725
CHG-12	6.51 (16)	7.23 (10)	11.51 (22)	68.67 (20)	0.23 (4)	0.42 (13)	94.58	5.42	8	1.61	1053	3761
CHG-21	4.59 (58)	7.39 (20)	11.47 (10)	70.98 (51)	0.21 (3)	0.36 (8)	95.02	4.98	7	1.36	972	3227
CHG-30	7.38 (35)	7.06 (20)	11.51 (6)	68.40 (22)	0.29 (2)	0.62 (2)	95.27	4.73	5	1.72	1323	5526
CHG-8	7.86 (53)	6.75 (48)	11.10 (17)	68.64 (60)	0.26 (1)	0.48 (4)	95.12	4.88	5	1.82	1203	4285
CHT-3	7.70 (25)	6.53 (16)	11.65 (10)	71.54 (20)	0.80 (5)	1.76 (5)	95.02	–	17	1.69	3477	15 008

Note: Values in parentheses are one standard deviation of *n* replicate analyses. AI = (Na + K)/Al in molar fraction.

peralkaline melts is primarily due to the enhanced solubility of the molecular species. Monazite will dissolve in a granitic melt until the melt has reached the saturation level of dissolved component; however, the dissolved cations such as REE³⁺ can only be dissolved if the suitable anions are available for charge balance. The incorporation of rare-Earth elements in silicate melts increases with increasing melt depolymerisation, being dependent on the availability of highly coordinated sites (Flynn & Burnham, 1978; Ponader & Brown, 1989). Peralkaline melts are characterized by a higher number of sites coordinated by non-bridging oxygen, implying lower degree of polymerization compared to peraluminous melts. Therefore, peralkaline melts can incorporate higher La contents than peraluminous melts. The preference of La for peralkaline melts is also supported by the similarity

among the ionic radius of La (117 pm) and Na (116 pm). Similar trends have been observed also for fluid-melt partition coefficients of La, with higher coefficient values in peraluminous melts than in peralkaline melts (Borchert *et al.*, 2010). The addition of alkalis could deform the structure of the melt via depolymerization, thus increasing the availability of non-bridging oxygens and thus La-monazite solubility. It is not clear why monazite solubility should increase slightly in peraluminous melts.

3.3. The effect of temperature on monazite solubility

The influence of temperature on monazite solubility is evident when comparing solubility data from melts at 1400 °C with those at lower temperatures and similar AI

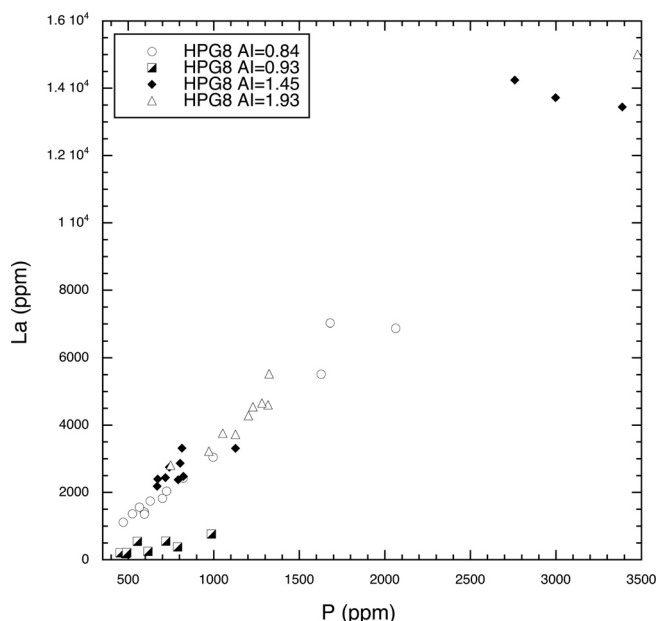


Fig. 1. Relationship between the solubility of La-monazite and phosphorous content in haplogranitic melts.

(Fig. 2). At 1400 °C, monazite solubility in peralkaline melts may exceed by up to three times the solubility measured at lower temperatures (near 800 °C). Considering the peraluminous melts, this difference increases up to five times.

The solubility data are plotted as function of $10\,000/T$ to constrain the temperature dependence of La-monazite solubility for the four series of starting materials (Fig. 3). La-monazite solubility increases by about 5–6 times over the temperature range from 750 to 1400 °C. The temperature dependence of La-monazite solubility (in ppm by weight, or mg/kg) for the four melt compositions can be described by the following equations:

$$\text{Log(La)} = 4.89 \pm 0.21 - (0.18 \pm 0.02)(10\,000/T) \quad \text{for } 750 < T < 1400\text{ °C; AI} = 0.84 \quad (11)$$

$$\text{Log(La)} = 7.33 \pm 0.87 - (0.51 \pm 0.09)(10\,000/T) \quad \text{for } 720 < T < 850\text{ °C; AI} = 0.93 \quad (12)$$

$$\text{Log(La)} = 5.48 \pm 0.11 - (0.23 \pm 0.01)(10\,000/T) \quad \text{for } 750 < T < 1400\text{ °C; AI} = 1.45 \quad (13)$$

$$\text{Log(La)} = 5.33 \pm 0.19 - (0.19 \pm 0.02)(10\,000/T) \quad \text{for } 750 < T < 1400\text{ °C; AI} = 1.93 \quad (14)$$

where T is temperature in K and the constants have been derived by linear regression. The similar constants imply that the temperature dependence of La-monazite solubility is comparable for the melt compositions with AI = 0.84, 1.45, and 1.93. The different constants for the melt

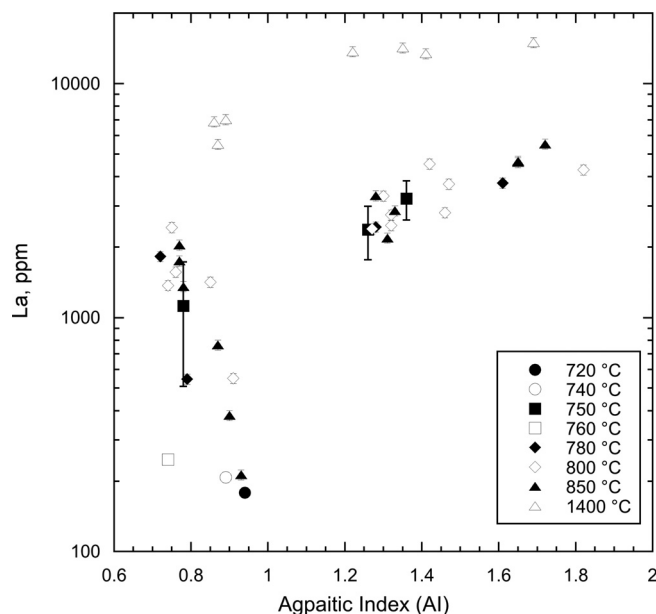


Fig. 2. The effect of the melt apgaitic index on La-monazite solubility. Error bars indicate a standard error of $\pm 5\%$ of the measured value. Below ~ 500 ppm, errors (1 sigma) are closer to 10–15% based on microprobe counting statistics.

composition with AI = 0.93 could be explained by the absence of solubility data for temperatures above 850 °C.

4. Discussion

4.1. Equilibrium consideration and effect of pressure

Attainment of equilibrium is checked by runs with different melt composition and by approaching equilibrium from dissolution experiments. Even though reversal experiments (crystallization experiments) were not performed in this study, the experimental data were compared to the literature data which indicate the attainment of equilibrium with direct and reverse experiments for experimental durations and temperatures similar to those employed in this study.

Run duration necessary for the attainment of equilibrium was evaluated by comparing our experimental data with previous studies done by Montel (1986, 1993), taking into account both dissolution and crystallization experiments. Unfortunately, run times are not reported in these studies, so we compare our solubility data with previous studies at the same experimental condition, because similar results would support a close approach to equilibrium. The measured solubility of monazite in water-saturated haplogranite melt at 800 °C is similar or slightly higher than the result of Montel (1986, 1993), suggesting a relatively close approach to equilibrium (Fig. 4).

However, at 800 °C, 2 kbar, Duc-Tin & Keppler (2015) used synthetic La monazite in the same HPG8 melt composition, but failed to reach equilibrium even with run duration of 1–2 months for the metaluminous to peraluminous compositions and all the La monazite solubility data are very

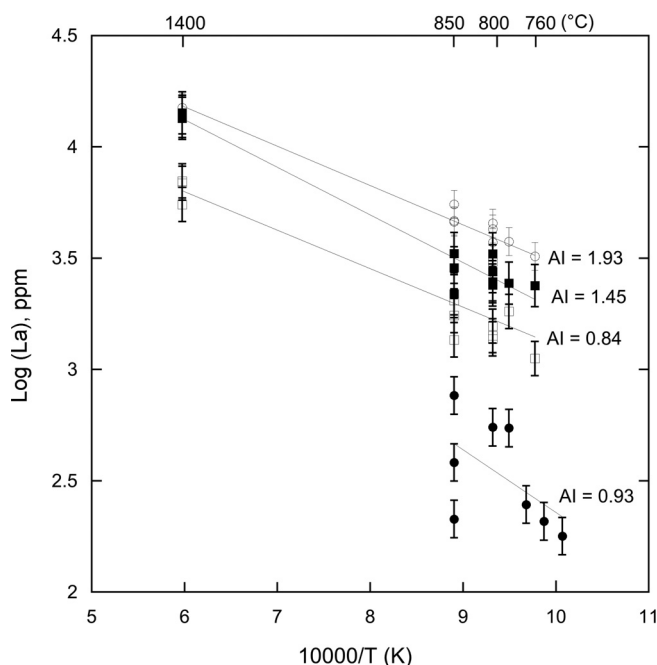


Fig. 3. Dependence of La-monazite solubility on temperature. Shown are data and linear fits from hydrous glasses obtained from the four starting glasses with AI = 1.93 ($R = 0.96$), AI = 1.45 ($R = 0.98$), AI = 0.93 ($R = 0.91$), AI = 0.84 ($R = 0.92$). Error bars indicate a standard error of $\pm 5\%$ of the measured value.

low compared to the present study and Montel (1986, 1993). This might be due to the crystallinity of the synthetic La-Monazite used by Duc-Tin & Keppler (2015). The material we used was very fine grained and apparently reacted quickly with the HPG8-based melts. In addition the starting haplogranitic glass compositions Duc-Tin & Keppler (2015) used were melted only 1–2 h at 1600 °C. The time is too short for complete homogenization, based on our experience. For our HPG8 based melt composition, we have melted the peraluminous composition at 1600 °C several weeks to obtain a homogenized crystal-free melt. This may be the reason why we measure higher solubility in homogeneous melt at shorter duration.

In solubility experiments, in general, it is easier to get low solubility values because of not reaching equilibrium (*e.g.*, slow monazite dissolution), but it is difficult to get higher solubility as observed in our results; this suggests that our results indicate relatively close approach to the equilibrium.

According to Linnen (2005), accessory-phase solubilities are not affected by the water content of peralkaline melts and, for subaluminous melts, water contents above 2 wt% yield nearly constant solubilities; Linnen (2005) concluded that water does not exert significant control on accessory-phase solubility. Moreover, the study of Rapp & Watson (1986) and the data of Montel (1986, 1993) suggest that pressure has little, if any, effect on monazite solubility, in spite of very different experimental techniques and natural and synthetic melt composition plus different LREE monazite [(La, Ce, Sm)PO₄] used; our experimental data agree remarkably well and also indicate that pressure has very little effect on monazite solubility (note that recent work of

Skora & Blundy (2012) shows this is not true at 30 kbar pressure).

4.2. Comparison with previous solubility studies in silicate melts

Solubility of CePO₄ and SmPO₄ at 800 °C and 2 kbar, under water-saturated conditions, in the model system SiO₂–Al₂O₃–Na₂O–K₂O are similar to our results (Fig. 5; Montel, 1986, 1993). Comparison of solubility data for monazite having compositions close to pure LREEPO₄ is feasible given that all the LREE-phosphates (La–Sm) have similar solubility (Montel, 1993).

Rapp & Watson (1986) estimated monazite solubility in dissolution experiments of natural crystals in a hydrous (1–6 wt% H₂O) natural granitic melt (AI = 0.86) at 8 kbar over the temperature range 1000–1400 °C using diffusion profiles of LREE in the glass adjacent to the dissolving crystals. Solubility data corrected to LREE activities for the effect of U, Th, and Y ($X_{mnz}^{LREE} = 0.96$; Stepanov *et al.*, 2012), give values varying from 1400 ppm to 2 wt% with highest solubilities observed at the highest temperatures investigated. The lack of published glass compositions and the different experimental conditions make a direct comparison with our data difficult. However, we note that using the algaic index of the starting material from Rapp & Watson (1986), their LREE solubility data at the highest temperatures are comparable to our values at 1400 °C (Fig. 5). Differences with our data set can be explained by the presence of elements such as Ca, Fe, and Mg causing an increase of monazite solubility.

The LREE concentrations of monazite-saturated peraluminous natural liquids have been investigated under water-saturated conditions between 850 and 1000 °C at 8 kbar by Rapp *et al.* (1987). They observed LREE concentrations in melts slightly higher than our data, perhaps reflecting the effect of higher pressure (Fig. 5). However, the only study that systematically investigated the effect of pressure on monazite solubility indicates a negative dependence between these parameters (Stepanov *et al.*, 2012). More probably this difference could be explained again by the presence of Ca and of other cations (*e.g.* Fe, Mg) in the natural composition, enhancing monazite solubility.

The solubility of synthetic pure REE phosphates in water-saturated haplogranitic silicate liquids has been measured at 2 kbar and 800–1100 °C by Keppler (1993) and Duc-Tin & Keppler (2015). Results of Keppler (1993) show that La-monazite solubility, in melts with AI = 1 is in the range 340–770 ppm, similar to our results for melts with AI varying from 0.87 to 0.93 (Fig. 5).

The study of Duc-Tin & Keppler (2015) shows that REE phosphates solubility increase slightly with increasing AI for the peralkaline melts, and for the peraluminous melts a weak increase with decreasing the AI is observed, as in our study. However, the solubility of La-monazite appears generally low (140–400 ppm, Figs. 5 and 6). Despite the long run durations (up to 2 months), the low La solubility can be due to the grain size (30–50 μ m) of the crystals used for

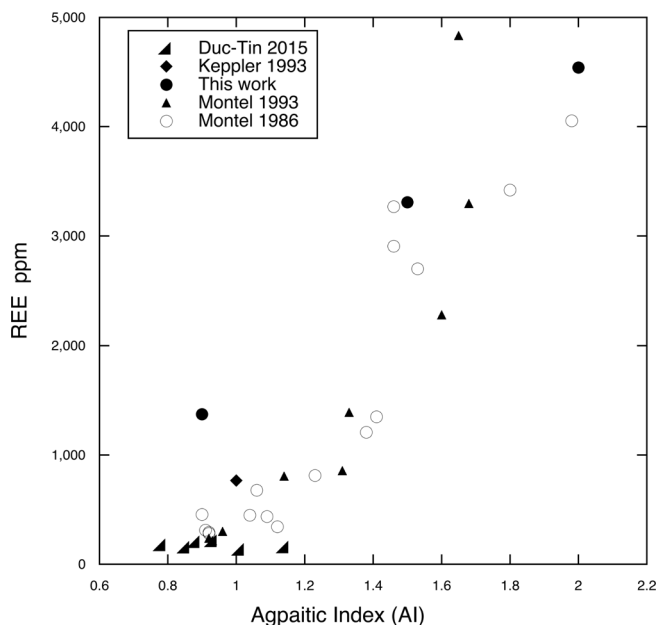


Fig. 4. Comparison of monazite solubility at 800 °C and 2 kbar, between this work and previous studies of Montel (1986, 1993), Duc-Tin & Keppler (2015) and Keppler (1993).

the experiments and to glass inhomogeneity, as testified by the presence of diffusion profiles around crystals. The absence of clear La solubility dependence with melt peralkalinity might be explained by the narrow range of peralkalinity (1.01–1.37) explored, as well as analytical errors at low La abundances.

Corrected LREE concentrations in aluminous granitic liquids, produced by melting of subducted sediments, coexisting with monazite at 800–1000 °C and 35–45 kbar are comparable to the results of this study for melts with AI in the range 0.79–0.94 (Fig. 6; Hermann & Rubatto, 2009). Hydrous melting of sediments at 700–1250 °C and 30 kbar leaves monazite as residual phase, thus permitting an estimation of the solubility of LREE (Skora & Blundy, 2010, 2012), although the amount of dissociation of La^{3+} and $(\text{PO}_4)^{3-}$ in the melt at high pressures makes difficult a direct comparison with the more abundant low-pressure data (as discussed in Skora & Blundy, 2012).

Corrected LREE concentrations in these broadly granitic silicate liquids show that monazite solubility varies from 80 to 2700 ppm, with the higher values in experiments doped with stoichiometric LREE/P ratios. Solubility data obtained in experiments with stoichiometric doping are comparable to our results with similar agpaitic index (Fig. 6).

Monazite solubility in synthetic granitic composition at 750–1200 °C, 10–50 kbar, and 10 wt% H_2O added was determined by Stepanov *et al.* (2012). Their corrected LREE concentrations are comparable to our solubility data for peraluminous melts with AI in the range 0.69–0.91 (Fig. 6). Higher solubility for $\text{AI} < 0.69$ are probably attributable to the higher water and calcium content of the melt in their experiments.

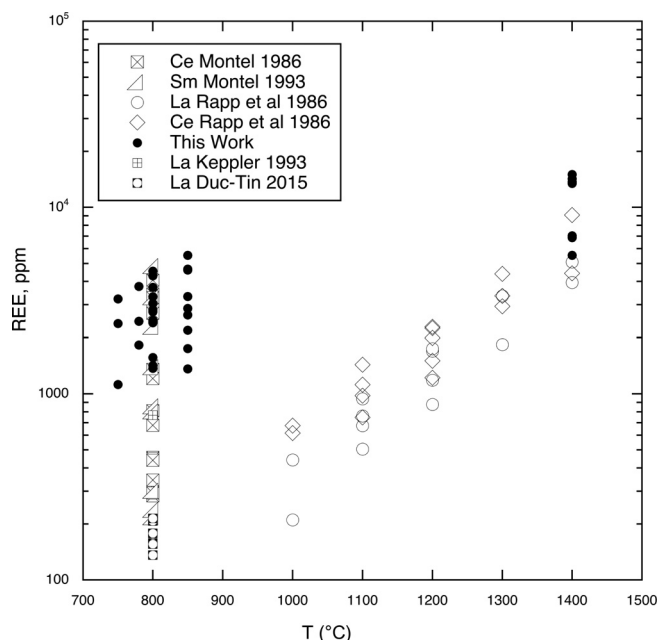


Fig. 5. Comparison of monazite solubility change with temperature between this study and Montel (1986, 1993), Rapp & Watson (1986), Keppler (1993) and Duc-Tin & Keppler (2015).

In the frame of a Th and U partitioning study between monazite and water-saturated $\text{SiO}_2\text{--Al}_2\text{O}_3\text{--Na}_2\text{O}$ melts, the solubility of Ce was measured at 10 kbar and 900–1100 °C (Fig. 6; Xing *et al.*, 2013). Corrected Ce concentrations are comparable to our results considering the different compositions and P - T conditions.

4.3. New expression for monazite solubility

In order to constrain monazite saturation in a wide range of compositions from peraluminous to peralkaline, our data have been combined with those obtained from model (*i.e.* $\text{SiO}_2\text{--Al}_2\text{O}_3\text{--Na}_2\text{O--K}_2\text{O}$) and natural systems with $\text{CaO} + \text{FeO} + \text{MgO} < 3 \text{ wt\%}$ (Montel 1986, 1993; Rapp & Watson, 1986; Hermann & Rubatto, 2009; Skora & Blundy, 2010, 2012; Stepanov *et al.*, 2012; Duc-Tin & Keppler, 2015). Data from the study of Xing *et al.* (2013) were not used for the calibration given that the absence of K_2O and CaO leads to extreme values of the compositional melt parameter M , defined as the atomic fractions of $(\text{Na} + \text{K} + 2\text{Ca}) \times \text{Al}^{-1} \times (\text{Al} + \text{Si})^{-1}$, both for peraluminous and peralkaline compositions when compared to the melt compositions of the database; *i.e.* for peraluminous compositions $M_{\text{Xing}} = 0.38\text{--}0.66$ and $M_{\text{database}} = 0.80\text{--}1.41$, for peralkaline composition $M_{\text{Xing}} = 0.79\text{--}1.09$ and $M_{\text{database}} = 1.17\text{--}2.48$. For the new calibration, multiple linear regression was applied to the whole data set composed of 43 new experiments from this study, and 135 published experimental runs in which melts are buffered by monazite. To consider the role of Th, U, and Y in monazite, LREE concentrations were corrected to LREE activities by assuming $a(\text{LREE})_{\text{mnz}} = X_{\text{mnz}}^{\text{LREE}} = \text{LREE}_{\text{mnz}} / (\text{LREE}_{\text{mnz}} + \text{Y}_{\text{mnz}} + \text{Th}_{\text{mnz}} + \text{U}_{\text{mnz}})$.

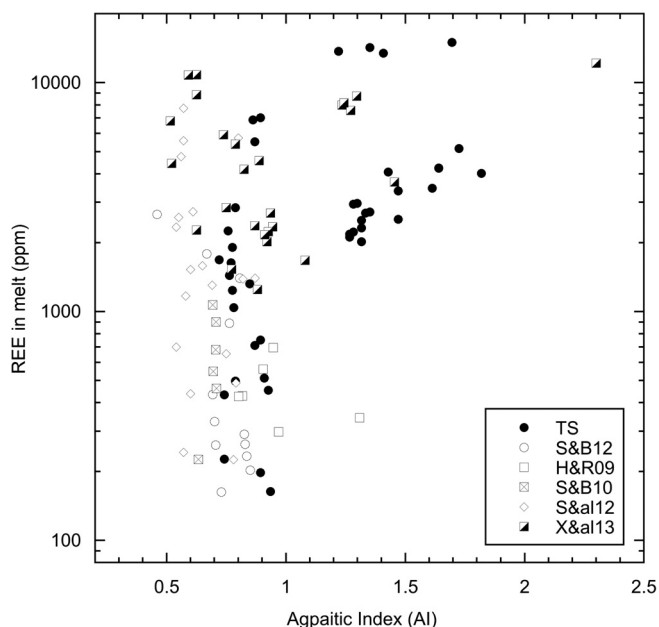


Fig. 6. Comparison of monazite solubility change with the agpaic index. LREE (La + Ce + Pr + Nd + Sm) is used to represent the monazite solubility corrected for the effect of Th, U, and Y (see text). TS, this study; H&R09, Hermann & Rubatto (2009); S&B10, Skora & Blundy (2010); S&B12, Skora & Blundy (2012); S&al12, Stepanov *et al.* (2012); X&al13, Xing *et al.* (2013).

The total of 178 experimental runs used for the calibration of monazite solubility in melts cover the following range: 720–1400 °C, 0.001–45 kbar, up to 29 wt% H₂O, and 0.7–2.5 *M*. The best fit resulted in the following equation:

$$\ln \Sigma \text{LREE} = 12.77(\pm 0.49) + 1.52(\pm 0.15)M + 0.44(\pm 0.10)(\text{H}_2\text{O})^{0.5} - 9934(\pm 632)/T - 36.79(\pm 6.15)P/T + \ln X_{\text{mnz}}^{\text{LREE}} \quad (15)$$

where ΣLREE is the sum of La–Sm in ppm, *M* is the dimensionless melt compositional parameter $(\text{Na} + \text{K} + 2\text{Ca}) \times \text{Al}^{-1} \times (\text{Al} + \text{Si})^{-1}$, H₂O is in weight percent, *T* is in K, *P* in kbar, $X_{\text{mnz}}^{\text{LREE}}$ is the mole fraction of LREE in monazite $\text{LREE}_{\text{mnz}}/(\text{LREE}_{\text{mnz}} + \text{Y}_{\text{mnz}} + \text{Th}_{\text{mnz}} + \text{U}_{\text{mnz}})$. One-standard-deviation uncertainties on parameters are given in parentheses. Based on available data, we suggest that water content does not have a large effect on monazite solubility. The compositional parameter *M* is able to describe the effect of melt composition on monazite solubility for peraluminous and peralkaline compositions. The proposed equation reproduces 76% and >95% of the data within an uncertainty of $\pm 10\%$ and $\pm 20\%$, respectively (Fig. 7). Caution should be taken if the equation is used on Ca-, Fe-, Mg-rich melts (*i.e.* CaO + FeO + MgO > 3 wt%).

Figure 8 graphically compares different calibrations of monazite solubility at *M* = 1, H₂O = 6 wt%, and *P* = 2 kbar (Rapp *et al.*, 1987; Montel, 1993; Stepanov *et al.*, 2012; this study). The new proposed calibration is in good agreement

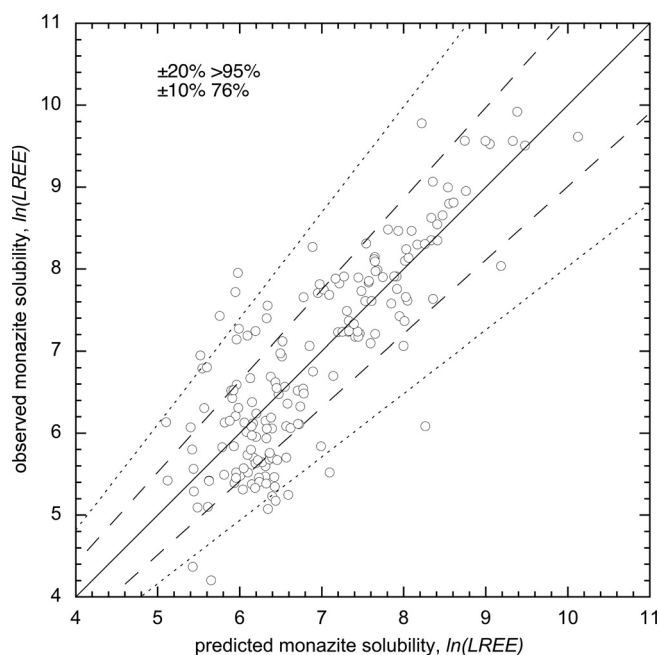


Fig. 7. Measured monazite solubility as a function of predicted monazite solubility using Eq. (15). The solid line represents equality between observed and estimated monazite solubility. Dashed and dotted lines represent, respectively, $\pm 10\%$ and $\pm 20\%$ deviation from equality.

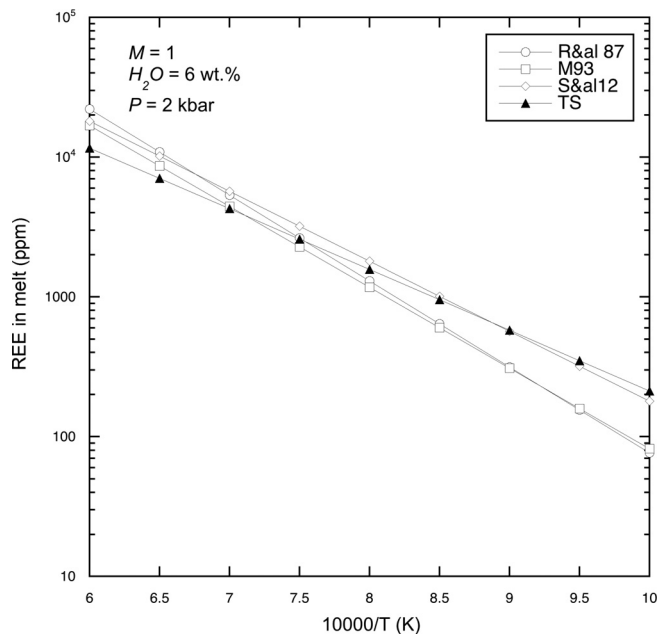


Fig. 8. New calibration of monazite solubility in melts and its comparison with previous calibrations. Arrhenius plot for total LREE solubility in melts. Lines of monazite solubility at *M* = 1, H₂O = 6 wt%, *P* = 2 kbar calculated from Eq. (15), TS (TS: this study) and from equations by Rapp *et al.* (1987), R&al87; Montel (1993), M93; and Stepanov *et al.* (2012), S&al12.

with the calibration proposed by Stepanov *et al.* (2012), given that solubility equations are based on the same parameters. Differences between these two calibrations are

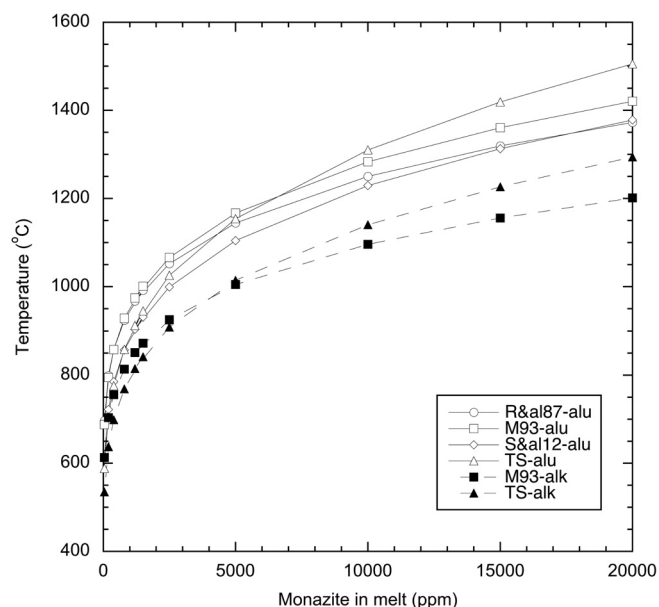


Fig. 9. Comparison of our new monazite solubility model (TS, this study) with models of Rapp *et al.* (1987), R&al87; Montel (1993), M93; and Stepanov *et al.* (2012), S&al12. Solid lines with open symbols are the variations of temperature and LREE concentration in peraluminous compositions (constant $M = 0.9$) and the dashed lines with filled symbols for peralkaline compositions (constant $M = 1.4$).

ascribed to our extended dataset considering peralkaline melts whereas those of Stepanov *et al.* (2012) was limited to peraluminous and metaluminous melts. Calibrations by Rapp *et al.* (1987) and Montel (1993) are also in good agreement with Stepanov *et al.* (2012) and our proposed calibration for temperatures >950 °C, whereas differences tend to increase at lower temperatures. Possibly, the effect of pressure that was not taken into account by Rapp *et al.* (1987) and Montel (1993) is more evident at lower temperatures, increasing thus the differences between the calibrations. Our calibration for monazite solubility predicts monazite saturation at 4400 ppm LREE in melt at 1100 °C, 2 kbar, $M = 1$ and 6 wt% H_2O . Under the same conditions monazite saturations are predicted at 3600, 4000, 4800 ppm LREE, respectively, by the equation of Rapp *et al.* (1987), Montel (1993), and Stepanov *et al.* (2012). At lower temperature our equation predicts 870 ppm LREE at 850 °C, 2 kbar, $M = 1$ and 6 wt% H_2O , in good agreement with the 750 ppm LREE predicted by the calibration of Stepanov *et al.* (2012), whereas 360 and 460 ppm LREE are predicted respectively by the calibrations of Rapp *et al.* (1987) and Montel (1993).

In Fig. 9 we can compare our new model to the previous ones for peraluminous ($M = 0.9$) and peralkaline melts ($M = 1.4$). For peraluminous melts our model show no appreciable differences below 900 °C with the calibration of Stepanov *et al.* (2012), whereas the differences at higher temperatures could be ascribed to the limited compositional variation used in their calibration. Also, differences with the calibrations of Rapp *et al.* (1987) and Montel (1993) could be due to the compositional ranges used in these

calibrations. For peralkaline compositions, our model agrees well with Montel (1993) in the range 900–1000 °C and tend to diverge from it at lower and higher temperatures. The lack of the compositional parameter in the models of Rapp *et al.* (1987) and Stepanov *et al.* (2012) prevents a comparison with our calibration for peralkaline melts. Nevertheless, the above discussion shows how our calibration, given the extended dataset used, is suitable for predicting monazite saturation from peraluminous to peralkaline melts taking into account all the parameters influencing monazite solubility.

5. Conclusions

We present new experimental results on La-monazite saturation in a variety of different melt compositions at high temperatures and at atmospheric and crustal pressures. From the combination of our experimental results with those of previous studies, we developed an empirical model which links the LREE concentrations of melts at monazite saturation to a compositional parameter M $[(Na + K + 2Ca) \times Al^{-1} \times (Al + Si)^{-1}]$, water content, temperature, and pressure. The new monazite saturation model is relatively straightforward to use and is valid for a wide range of compositions. The model includes all relevant chemical and physical parameters, including the compositional parameter that was not present in some of the previous models.

In summary, our new model agrees well with previous studies in peraluminous, metaluminous, and peralkaline compositions reproducing, respectively, 76% and $>95\%$ of the data within an uncertainty of $\pm 10\%$ and $\pm 20\%$. Thus, it can be applied from peraluminous to peralkaline melt compositions. The limitation of our model is represented by its applicability to FeO-, CaO-, MgO-rich compositions (*i.e.* >3 wt%); future studies should address the effect of these chemical parameters on monazite solubility.

Acknowledgements: We thank Raul Carampin at CNR Padova for his invaluable help with the microprobe analysis. Reviews by Peter Tropper and anonymous reviewer considerably improved an earlier draft of this article. Associated editor Roland Stalder is thanked for his detailed comments and suggestions. MRC acknowledges support from MIUR (PRIN-2009PZ47NA). EP acknowledges support from project FAR2014 and EU-Life-Ecotiles (LIFE14 ENV/IT/000801). AF acknowledges support from the GACR (Project 18-01982S). DBD acknowledges the support of an ERC Advanced Grant (EVOKES).

References

- Bea, F. (1996): Residence of REE, Y, Th and U in granites and crustal protoliths; implications for the chemistry of crustal melts. *J. Petrol.*, **37**, 521–552.
- Borchert, M., Wilke, M., Schmidt, C., Cauzid, J., Tucoulou, R. (2010): Partitioning of Ba, La, Yb and Y between haplogranitic melts and aqueous solutions: An experimental study. *Chem. Geol.*, **276**, 225–240.

- Chakhmouradian, A.R. & Wall, F. (2012): Rare earth elements: Minerals, Mines, Magnets (and more). *Elements*, **8**, 333–340.
- Chakhmouradian, A.R. & Zaitsev, A.N. (2012): Rare earth mineralization in igneous rocks: sources and processes. *Elements*, **8**, 347–353.
- Di Matteo, V., Carroll, M.R., Behrens, H., Vetere, F., Brooker, R. (2004): Water solubility in trachytic melts. *Chem. Geol.*, **213**, 187–196.
- Duc-Tin, Q. (2007): Experimental studies on the behaviour of rare earth elements and tin in granitic systems PhD thesis, University of Tübingen. <https://publikationen.uni-tuebingen.de/xmlui/handle/10900/49059>
- Duc-Tin, Q. & Keppler, H. (2015): Monazite and xenotime solubility in granitic melts and the origin of the lanthanide tetrad effect. *Contrib. Mineral. Petrol.*, **169**, 8, <https://doi.org/10.1007/s00410-014-1100-9>.
- Fabrizio, A. & Carroll, M.R. (2008): Experimental constraints on the differentiation process and pre-eruptive conditions in the magmatic system of Phlegraean Fields (Naples, Italy). *J. Volcanol. Geotherm. Res.*, **171**, 88–102.
- Fabrizio, A., Rouse, P.J., Carroll, M.R. (2006): New experimental data on biotite plus magnetite plus sanidine saturated phonolitic melts and application to the estimation of magmatic water fugacity. *Am. Mineral.*, **91**, 1863–1870.
- Flynn, R.T. & Burnham, C.W. (1978): An experimental determination of rare earth partition coefficients between a chloride containing vapor phase and silicate melts. *Geochim. Cosmochim. Acta*, **42**, 685–701.
- Hermann, J. & Rubatto, D. (2009): Accessory phase control on the trace element signature of sediment melts in subduction zones. *Chem. Geol.*, **265**, 512–526.
- Holtz, F., Behrens, H., Dingwell, D.B., Taylor, R.P. (1992): Water solubility in aluminosilicate melts of haplogranite composition at 2 kbar. *Chem. Geol.*, **96**, 289–302.
- Jones, A., Wall, F., Williams, C. (1996): Rare earth minerals: chemistry, origin and ore deposits, London, Chapman & Hall, 357 p.
- Keppler, H. (1993): Influence of fluorine on the enrichment of high field strength trace elements in granitic rocks. *Contrib. Mineral. Petrol.*, **114**, 479–488.
- Linnen, L.R. (2005): The effect of water on accessory phase solubility in subaluminous and peralkaline granitic melts. *Lithos*, **80**, 267–280.
- Lyakhovich, V. & Barinskii, R. (1961): Characteristics of the rare-earth assemblages in the accessory minerals of granitoids. *Geochemistry*, **6**, 495–509.
- Miller, C.F. & Mittlefehldt, D.W. (1982): Depletion of rare-earth elements in felsic magmas. *Geology*, **10**, 129–133.
- Montel, J.M. (1986): Experimental determination of the solubility of Ce-monazite in $\text{SiO}_2\text{-Al}_2\text{O}_3\text{-K}_2\text{O-Na}_2\text{O}$ melts at 800°C, 2 kbar, under H_2O -saturated conditions. *Geology*, **14**, 659–662.
- (1993): A model for monazite/melt equilibrium and application to the generation of granitic magmas. *Chem. Geol.*, **110**, 127–146.
- Overstreet, W. (1967): The geologic occurrence of monazite, USGS Professional Paper 530. U.S. Dept. of the Interior, Washington.
- Papale, P., Moretti, R., Barbato, D. (2006): The compositional dependence of the saturation surface of $\text{H}_2\text{O}+\text{CO}_2$ fluids in silicate melts. *Chem. Geol.*, **229**, 78–95.
- Poletti, J.E., Cottle, J.M., Hagen-Peter, G.A., Lackey, J.S. (2017): Petrochronological constraints on the origin of the Mountain Pass ultrapotassic and carbonatite intrusive suite, California. *J. Petrol.*, **57**, 1555–1598.
- Ponader, C.W. & Brown, C.E. (1989): Rare earth elements in silicate glass/melt systems: II. Interactions of La, Gd, and Yb with halogens. *Geochim. Cosmochim. Acta*, **53**, 2905–2914.
- Rapp, R.P. & Watson, E.B. (1986): Monazite solubility and dissolution kinetics: implications for the thorium and light rare-earth chemistry of felsic magmas. *Contrib. Mineral. Petrol.*, **94**, 304–316.
- Rapp, R.P., Ryerson, F.J., Miller, C.F. (1987): Experimental evidence bearing on the stability of monazite during crustal anatexis. *Geophys. Res. Lett.*, **14**, 307–310.
- Reed, S.J.B. (2005): Electron microprobe analysis and scanning electron microscopy in geology, 2nd edition, Cambridge University Press, 190 p.
- Seydoux, G.A., Wirth, R., Heinrich, W., Montel, J.M. (2002): Experimental determination of Thorium partitioning between monazite and xenotime using analytical electron microscopy and X-ray diffraction Rietveld analysis. *Eur. J. Mineral.*, **14**, 869–878.
- Skora, S. & Blundy, J. (2010): High-pressure hydrous phase relations of radiolarian clay and implications for the involvement of subducted sediment in arc magmatism. *J. Petrol.*, **51**, 2211–2243.
- , — (2012): Monazite solubility in hydrous silicic melts at high pressure conditions relevant to subduction zone metamorphism. *Earth Planet. Sci. Lett.*, **321–322**, 104–114.
- Spear, F.S. & Pyle, J.M. (2010): Theoretical modelling of monazite growth in a low-Ca metapelite. *Chem. Geol.*, **273**, 111–119.
- Stepanov, A.S., Hermann, J., Rubatto, D., Rapp, R.P. (2012): Experimental study of monazite/melt partitioning with implications for the REE, Th and U geochemistry of crustal rocks. *Chem. Geol.*, **300–301**, 200–220.
- Xing, L., Trail, D., Watson, E.B. (2013): Th and U partitioning between monazite and felsic melt. *Chem. Geol.*, **358**, 46–53.

Received 30 November 2017

Modified version received 28 May 2018

Accepted 4 June 2018